

Evaluation of Multimodal LLMs for Handwritten Receipt Information Extraction

1. Approach

The objective of this evaluation was to compare the performance of multimodal Large Language Models (LLMs) in extracting structured information from handwritten Indian receipts.

A dataset of **12 handwritten receipts** was used for evaluation. Each receipt was processed using three vision-language models:

- NVIDIA Nemotron Nano 12B VL through OpenRouter
- Command-A-Vision-07-2025 through Cohere
- Qwen3.6-27B through Groq

The models were evaluated on their ability to extract the following fields:

- Vendor name
- Date
- Amount
- Currency
- GST
- Invoice number

The extracted outputs from each model were converted into structured JSON format and compared against manually created ground truth values.

2. Evaluation Methodology

The evaluation framework compared model outputs against ground truth using field-level scoring.

Different comparison methods were applied depending on the field to handle formatting variations:

Field	Evaluation Method	Reason
Vendor	RapidFuzz partial ratio ≥ 90	Handles capitalization differences, abbreviations, and minor spelling variations.
Date	Normalize to YYYY-MM-DD format, then exact comparison	Handles different valid date representations.
Amount	Convert to numeric format, then exact comparison	Handles formatting differences such as decimal representation.
Currency	Normalize (₹, Rs, INR), then exact comparison	Treats different currency formats equally.

GST	Exact comparison	Prevents acceptance of hallucinated GST values.
Invoice Number	Remove leading zeros, then exact comparison	Ignores formatting differences.

Field accuracy was calculated as:

Accuracy = (Correct Predictions / Total Predictions) × 100

API cost was calculated using the total token usage reported by each provider:

Cost = (Total Tokens / 1,000,000) × Cost per Million Tokens

3. Accuracy and Cost Comparison

Accuracy Comparison

Model	Vendor	Date	Amount	Currency	Invoice	GST
Nemotron Nano 12B VL	91.67%	16.67%	75.00%	83.33%	83.33%	58.33%
Command-A-Vision-07-2025	100.00%	0.00%	50.00%	100.00%	91.67%	91.67%
Qwen3.6-27B	100.00%	50.00%	91.67%	83.33%	83.33%	91.67%

Cost Comparison

Model	Provider	Token Usage	Cost (12 Receipts)	Estimated Cost (100 Receipts)
Nemotron Nano 12B VL	OpenRouter	129k	\$0.03096	\$0.258
Command-A-Vision-07-2025	Cohere	121k	\$1.5125	\$12.604
Qwen3.6-27B	Groq	16.1k	\$0.01449	\$0.121

4. Final Recommendation

Based on the evaluation results, **Qwen3.6-27B provides the best overall cost-performance balance** among the evaluated models.

It achieved:

- Highest amount extraction accuracy (91.67%)
- Perfect vendor extraction accuracy (100%)

- Strong GST extraction accuracy (91.67%)
- Lowest estimated processing cost (\$0.121 per 100 receipts)

Based on these results, I would not necessarily use the same model or pipeline for both digital and handwritten documents. Digital invoices typically contain clean, machine-printed text and consistent layouts, making them well suited to OCR or document understanding models. Handwritten receipts present additional challenges such as varying handwriting styles, inconsistent formatting, and image quality, requiring models that are better equipped to interpret visual context.

The evaluation also highlights that **date extraction remains the most challenging field** across all three models, with the highest accuracy reaching only **50.00%**. This suggests that handwritten receipt extraction can benefit from additional pre-processing or post-processing, such as field normalization or validation rules, to improve reliability.

Overall, **Qwen3.6-27B is recommended for handwritten receipt extraction** due to its strong extraction performance and significantly lower processing cost.